EE2001 Electrical Circuits-I:

Continuation to:

Techniques of Circuit Analysis

23 October 2023

Outline

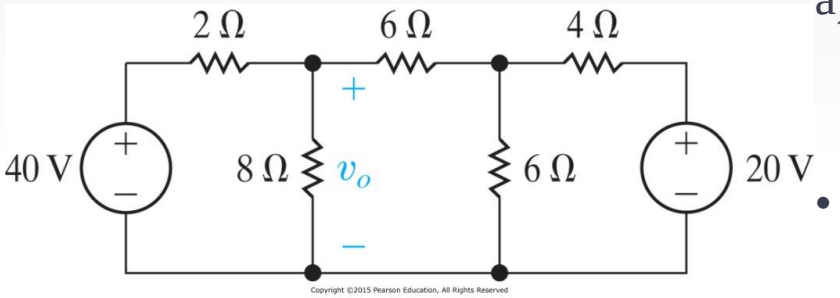


Continuation to: 2. Introduction to the Mesh-Current Method

3. The Superposition Principle

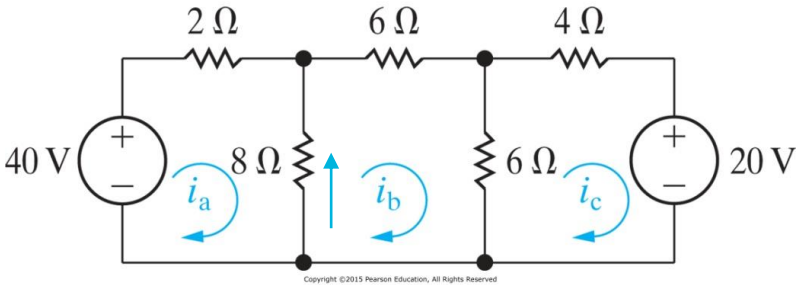
5 Introduction to the Mesh-Current Method

- Example-3



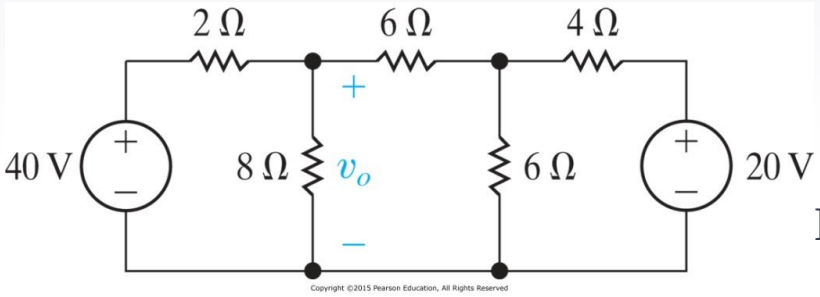
a) Use the **mesh-current method** to determine the **power associated with each voltage source** in the circuit.

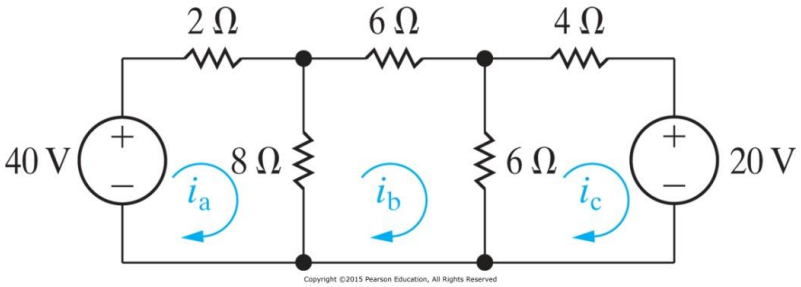
- There are **3 essential nodes** and **5 essential branches**. Therefore we must write [5 –



Introduction to the Mesh-Current Method

- Example-3





Organizing the equations

$$10i_a - 8i_b + 0i_c = 40$$

$$-8i_a + 20i_b - 6i_c = 0$$

$$0i_a - 6i_b + 10i_c = -20$$

Eliminating i_c from **equations 2 and 3**:

$$(Multiply\ with\ 5) \quad -8i_a + 20i_b - 6i_c = 0$$

$$(Multiply\ with\ 3) \quad 0i_a - 6i_b + 10i_c = -20$$

$$-40i_a + 100i_b - 30i_c = 0$$

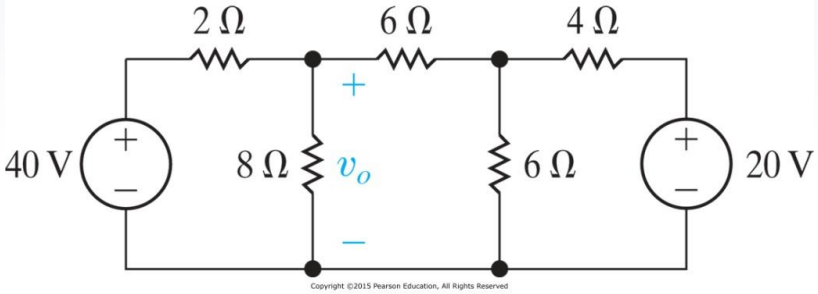
$$0i_a - 18i_b + 30i_c = -60$$

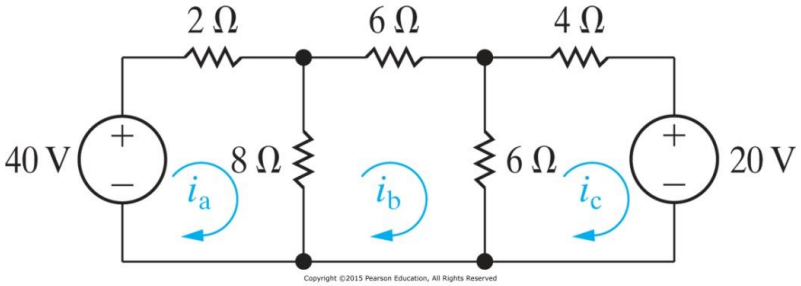
Adding the equations

$$-40i_a + 82i_b = -60$$

Introduction to the Mesh-Current Method

- Example-3





Using **the first original equation** and **the last derived equation** to **eliminate i_a,**

$$(Multiply\ with\ 4) \quad 10i_a - 8i_b = 40$$

$$-40i_a + 82i_b = -60$$

$$40i_a - 32i_b = 160$$

$$-40i_a + 82i_b = -60$$

We obtain

$$50i_b = 100$$

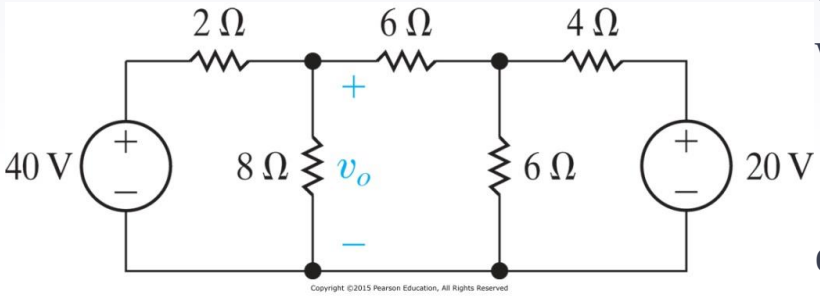
$$i_b = 2A$$

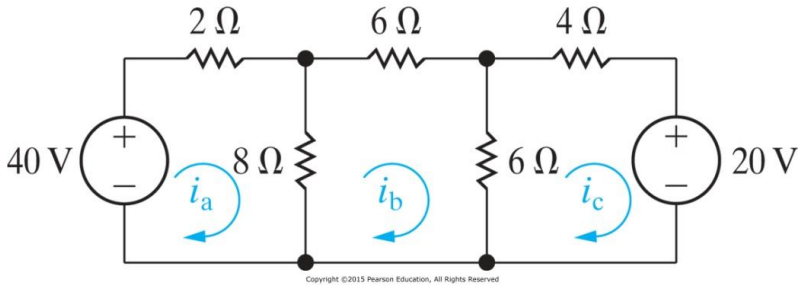
$$10i_a - 8(2) = 40 \qquad i_a = 5.6A$$

$$-6(2) + 10i_c = -20 \qquad i_c = -0.8A$$

5 Introduction to the Mesh-Current Method

- Example-3





The mesh current $\boldsymbol{i_a}$ **is identical with the branch current in the 40V source**, so the power associated with this source is

$$p_{40V} = -40i_a = -40(5.6) = -224W$$

The **minus sign means that this source is delivering power to the circuit**.

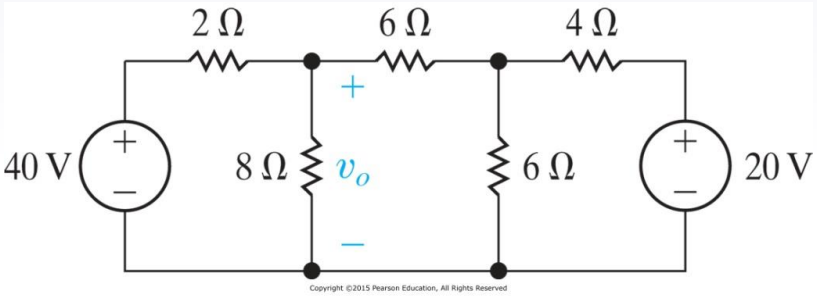
The mesh current $\boldsymbol{i_c}$ **is identical with the branch current in the 20V source**, so the power associated with this source is

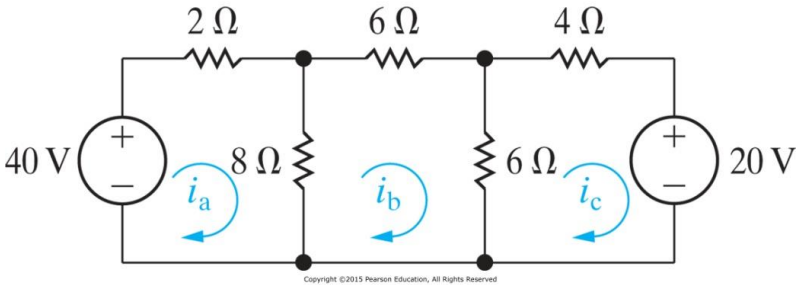
$$p_{20V} = 20i_c = 20(-0.8) = -16W$$

The 20 V source also **is delivering power to the network**.

Introduction to the Mesh-Current Method

- Example-3





b) Calculate **the voltage** v_o across the **8 Ω resistor**

The branch current in the 8 Ω resistor **in the direction of voltage drop** $\boldsymbol{v_o}$ is $\boldsymbol{i_a - i_b}$. Therefore

$$v_o = 8(i_a - i_b) = 8(5.6 - 2) = 8(3.6) = 28.8\ V$$

9 The Superposition Principle

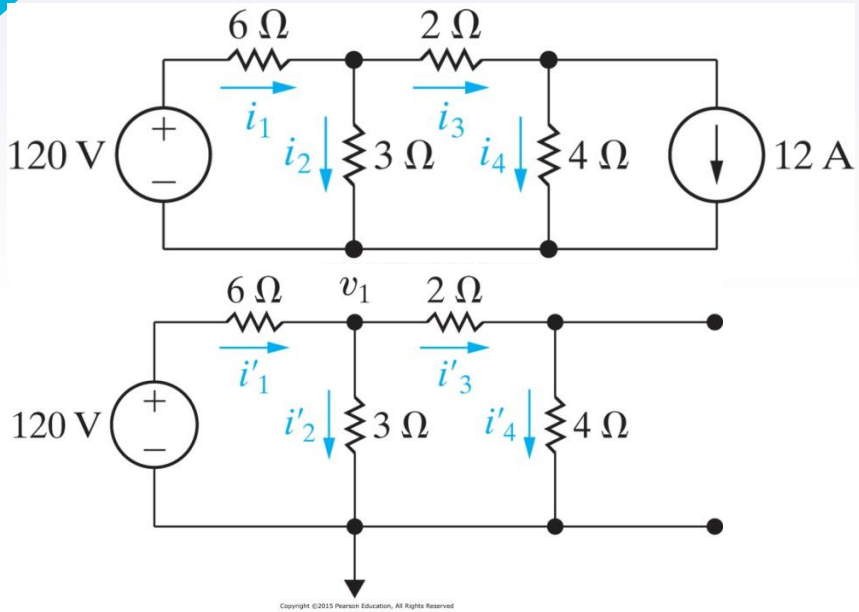
- The **superposition** principle is an important property of linear systems (including linear circuits).
- It states that whenever a linear system/circuit is **exited or driven by more than one independent sources of energy**, the resultant **total response is the sum of the individual responses** due to **each independent source acting alone**.
- This principle can be used in circuit analysis (**as long as the circuit is composed of linear elements** such as ideal resistors, capacitors, inductors, and current and voltage sources).
- **At the moment we will discuss only resistive networks/circuits**. However, the principle can be applied **to any linear circuit**.

The Superposition Principle

- The **superposition** principle can be used for both analysis and design of circuits – e.g., to design a circuit to **achieve a response composed of a sum of two elements/terms**.
- In case the independent sources which are driving the circuit **are fundamentally different**, it is needed to use superposition. **Otherwise**, the analysis may be complicated.
- The superposition principle can be applied by **considering one** independent source **at a time**, **deactivating the remaining** sources, and **calculating** the response
  - **Deactivating a current source** is equivalent to **replacing** it with **an OPEN CIRCUIT**.
  - **Deactivating a voltage source** is equivalent to **replacing** it with **a SHORT CIRCUIT**.

The Superposition Principle

- Example:

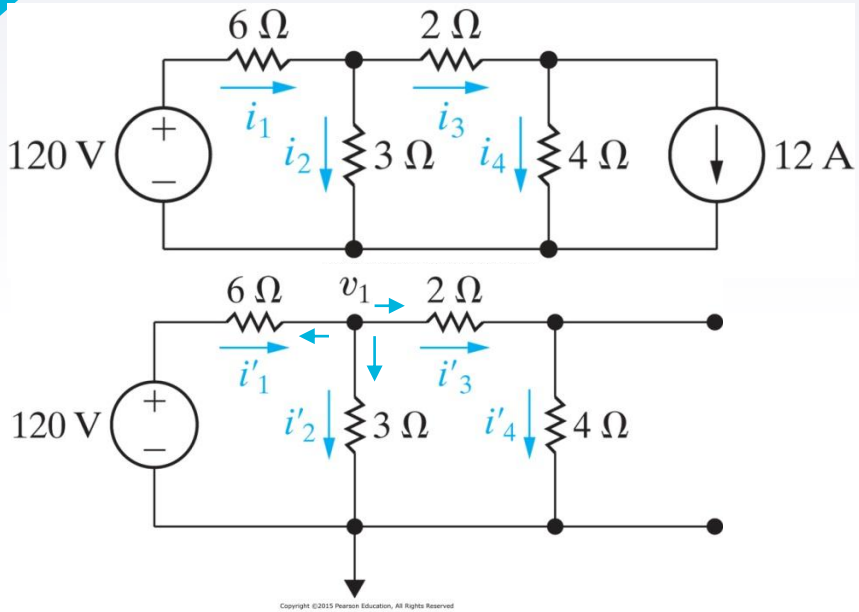


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- We begin by **finding the branch currents resulting from** 120 V **voltage source**.
- We denote those currents with **a prime (′)**.
- **Replacing** the **ideal current source** with **an open circuit deactivates** it. In this case, the branch currents are the **result of only the voltage source**.
- We can easily find the branch currents once we know the voltage across 3 Ω resistor, denoted as v_1.
- Considering the modified circuit has **2 essential nodes**, $(n_e - 1)$=(2-1)=1 **node-voltage equation** is needed to solve the circuit.

The Superposition Principle

- Example:



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- Choosing the lower essential node as reference, writing the **node-voltage equation** for the upper essential node,

$$\frac{v_1 - 120}{6} + \frac{v_1}{3} + \frac{v_1}{2+4} = 0$$

which gives us

$$v_1 = 30\ V$$

Now, we can calculate the branch currents:

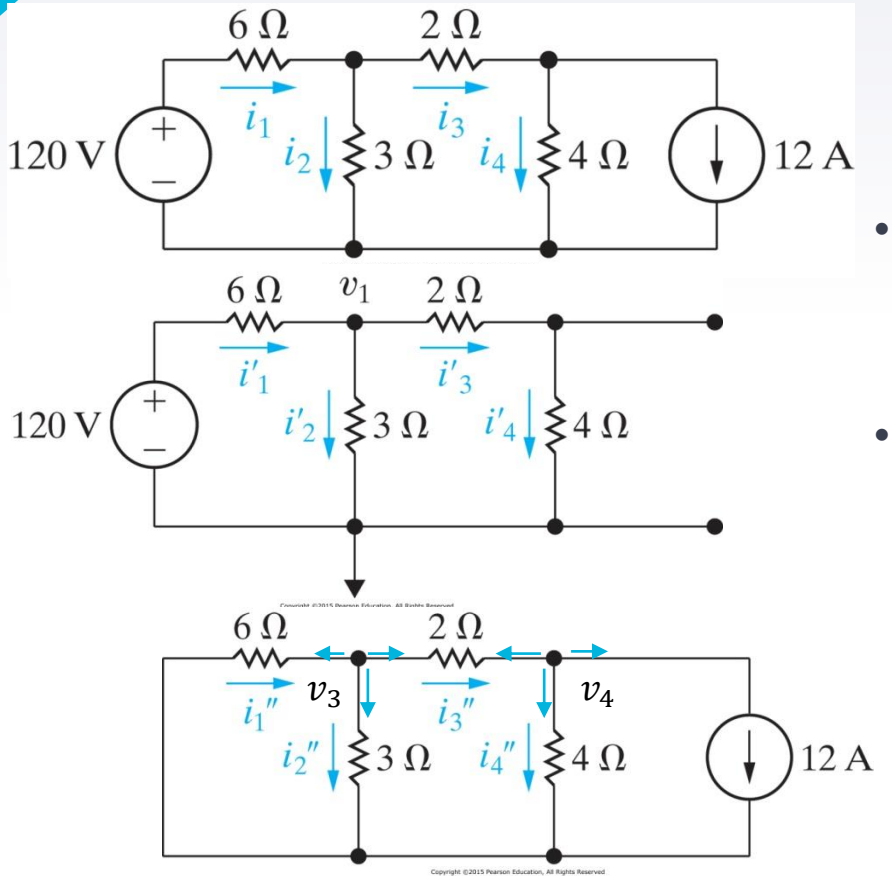
$$i'_1 = \frac{120 - 30}{6} = 15\ A$$

$$i'_2 = \frac{30}{3} = 10\ A$$

$$i'_3 = i'_4 = \frac{30}{6} = 5\ A$$

The Superposition Principle

- Example:



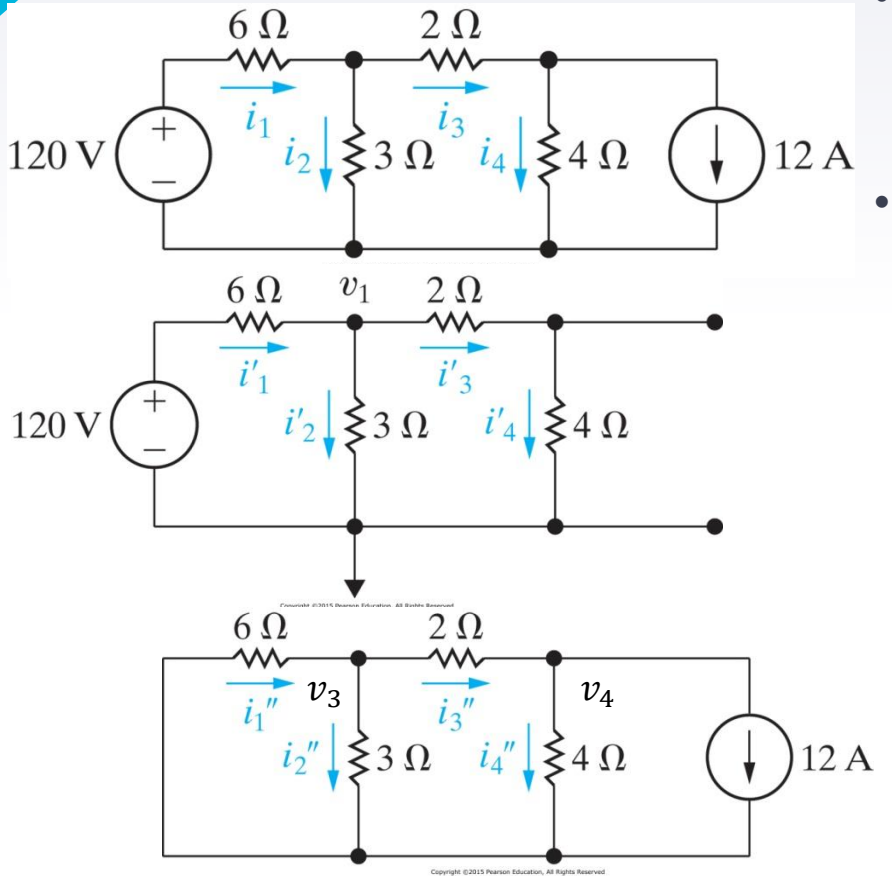
- To find the component of **branch currents resulting from the current source**, we **deactivate** the **ideal voltage source** by **short-circuiting it**.
- **The double-prime ($''$) notation** is used for the current components resulting from the ideal current source.
- In this modified circuit topology, there are **3 essential nodes**, so we need **2 node-voltage** equations.

$$\frac{v_3}{6} + \frac{v_3}{3} + \frac{v_3 - v_4}{2} = 0$$

$$\frac{v_4 - v_3}{2} + \frac{v_4}{4} + 12 = 0$$

The Superposition Principle

- Example:



- Solving the two equations with two unknowns,

$$v_3 = -12\ V$$
$$v_4 = -24\ V$$

- Now, we can write the branch currents for this modified circuit case

$$i_1'' = \frac{-v_3}{6} = \frac{12}{6} = 2\ A$$

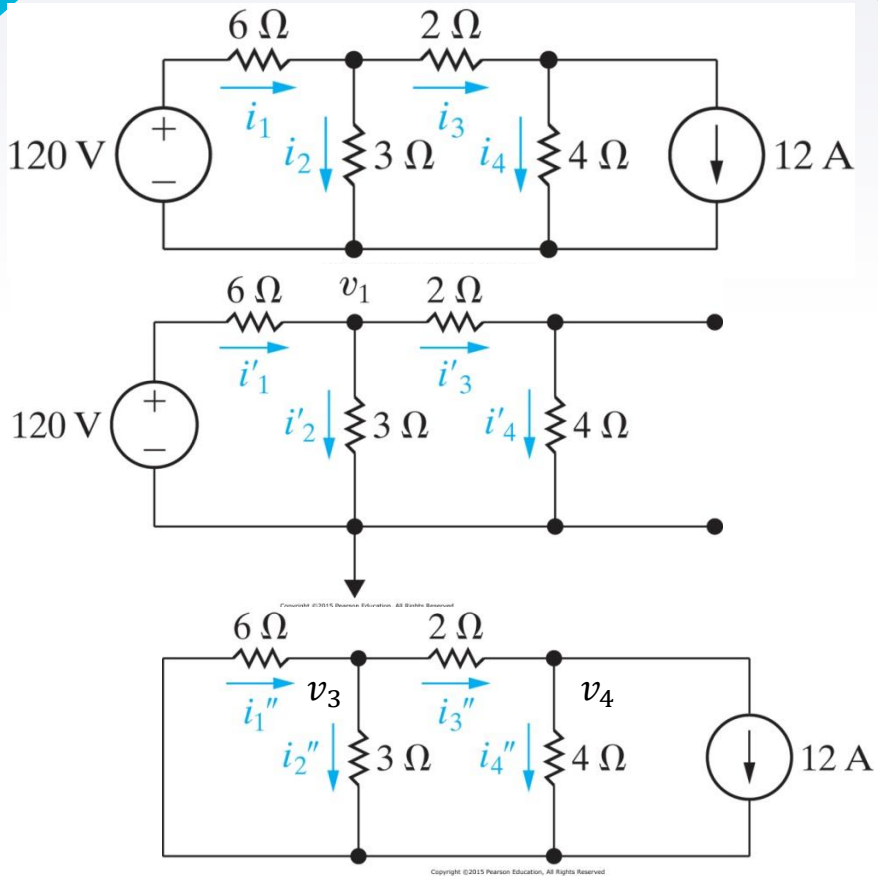
$$i_2'' = \frac{v_3}{3} = \frac{-12}{3} = -4\ A$$

$$i_3'' = \frac{v_3 - v_4}{2} = \frac{-12 + 24}{2} = 6\ A$$

$$i_4'' = \frac{v_4}{4} = \frac{-24}{4} = -6\ A$$

The Superposition Principle

- Example:



- The real currents in the original circuit are just the sum of the current components of the two considered cases:

$$i_1 = i_1' + i_1'' = 15 + 2 = 17\ A$$
$$i_2 = i_2' + i_2'' = 10 - 4 = 6\ A$$
$$i_3 = i_3' + i_3'' = 5 + 6 = 11\ A$$
$$i_4 = i_4' + i_4'' = 5 - 6 = -1\ A$$

End of the Lesson (23.10.23)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist